

Adriane M. McDonald

adrianemcdonald@ucsb.edu

(704) 582 9424

EDUCATION

6th year Ph.D. Candidate in Ecology, Evolution, and Marine Biology *2019 – Present*

Expected Graduation: Summer 2025

Advisor: Prof. Gretchen Hofmann

University of California, Santa Barbara

Santa Barbara, CA

B. S. in Biology, *Magna Cum Laude* *2015-2019*

Ethel Waddell Githii Honors Program

Spelman College

Atlanta, GA

FELLOWSHIPS AND GRANTS

NSF INTERN – Supplemental Grant

Awarded: July 2024

Funding Organization: The National Science Foundation

Collaborating Organization: NOAA Southeast Fisheries Science Center Marine Mammal and Sea Turtle Division

Amount Awarded: \$47,735.00

Research Accelerator Award

Awarded: May 2021

Funding Organization: Schmidt Family Foundation

Amount Awarded: \$8000.00

NSF Graduate Research Fellowship Program

Awarded: March 2021

Funding Organization: National Science Foundation

Center for Diverse Leadership in Science Early-Career Fellow

March 2021- Present

Funding Organization: UCLA Institute of the Environment & Sustainability

Coastal Fund Research Grant

Awarded: November 2019

Funding Organization: UCSB Associated Students and the Coastal Fund Board of Directors

Amount Awarded: \$973.00

Period of Grant Award: Dec. 2019-Dec. 2020

Title of Project: Tracking a marine heatwave, NEP19, along the Santa Barbara Coast

UC-HBCU Initiative Fellowship

September 2019 - Present

Funding Organization: The University of California Office of the President, the UCSB Graduate Division, and UCSB Ecology, Evolution, and Marine Biology Department

RESEARCH

Graduate Student Researcher

September 2019-Present

Ecology, Evolution, and Marine Biology, UC Santa Barbara

Committee: Prof. Gretchen Hofmann (Chair), Prof. Halley Froehlich, Prof. Nick Nidzieko, and Prof. Karen Chan.

Title: Investigating the Impacts of Marine Heatwaves on the Egg Quality and Early Stages of *Strongylocentrotus purpuratus*

Graduate Student Researcher

September 2020-August 2023

Ecology, Evolution, and Marine Biology, UC Santa Barbara

Collaborators: Jannine Chamorro, Ph.D. and Prof. Gretchen Hofmann

Title: Transgenerational plasticity as a mechanism of response to marine heatwaves in the purple sea urchin, *S. purpuratus*

Undergraduate Researcher

August 2017-May 2019

Biology, Spelman College

Principal Investigator: Dongfang Wang, Ph.D.

Title: Analyzing Promoter 5' Deletions on InvINH1 during Seed Development

Summer Student Fellow

May 2018-August 2018

Biology, Woods Hole Oceanographic Institution

Principal Investigator: Neel Aluru, Ph.D.

Title: Exposure to PCB153 affects Circadian Rhythm Behavior and Gene Expression in Zebrafish

Undergraduate Researcher

June 2017-August 2017

Ecology and Evolutionary Biology, UC Los Angeles

Collaborators: Shamou Bideau, Maria Gasca, Camille Gaynus

Principal Investigators: Paul Barber, Ph.D. and Peggy Fong, Ph.D.

Title: Urchins are Important Herbivores and Mediate Positive Effects of Sediment on Algal Turfs

PUBLICATIONS

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1. **McDonald, A.M.**, Hofmann, G. E. (2025). Marine heatwave temperatures alter maternal provisioning and early development in the purple sea urchin *Strongylocentrotus purpuratus* [Manuscript in preparation]. Department of Ecology, Evolution, and Marine Biology, University of California, Santa Barbara.
 2. Chan, K. Y. K., Kui, L., **McDonald, A. M.**, Ritger, A. L., & Hofmann, G. E. (2024). Coastal marine heatwaves in the Santa Barbara Channel: decadal trends and ecological implications. *Frontiers in Marine Science*, 11. <https://doi.org/10.3389/fmars.2024.1476542>

3. **McDonald, A. M.** (2023, July 5). *Land and water cover decreases likelihood of perceiving environmental problems, regardless of income*. National Science Foundation Long Term Ecological Research Network. <https://lternet.edu/stories/land-and-water-cover/>
4. Chamorro, J.D., **McDonald, A. M.**, Hofmann, G. H. (2023) Transgenerational plasticity as a mechanism of response to marine heatwaves in the purple sea urchin, *Strongylocentrotus purpuratus*, *Frontiers in Marine Science*, 10. <https://doi.org/10.3389/fmars.2023.1212781>
5. Aluru, N., Krick, K. S., **McDonald, A. M.** Karchner, S. I. (2020) Developmental Exposure to PCB153 (2,2',4,4',5,5'-Hexachlorobiphenyl) Alters Circadian Rhythms and the Expression of Clock and Metabolic Genes, *Toxicological Sciences*, 173(1), 41–52. <https://doi.org/10.1093/toxsci/kfz217>

TEACHING

Teaching Assistant, EEMB116 – Invertebrate Zoology	<i>Spring 2024</i>
Ecology, Evolution, and Marine Biology Department, UC Santa Barbara	
Head Teaching Assistant, EEMB2LL – Introductory Biology Lab	<i>Spring 2023</i>
Ecology, Evolution, and Marine Biology Department, UC Santa Barbara	
Teaching Assistant, MCDB1LL – Introductory Biology Lab	<i>Winter 2022</i>
Molecular, Cellular, and Developmental Biology Department, UC Santa Barbara	

SERVICE

Academic Research Consortium Graduate Mentor	<i>June-August 2023 and 2024</i>
Graduate Division, UC Santa Barbara	
FUERTE Mentor	<i>August 2020-August 2024</i>
Field-based Undergraduate Engagement through Research, Teaching, and Education, UC Santa Barbara	
Graduate Student Assistant	<i>September 2023-June 2024</i>
Office of Black Student Development, UC Santa Barbara	
LTER Graduate Student Writer	<i>August 2022 – August 2023</i>
Long-term Ecological Research Program (LTER) Network Office	
Lead Mentor for Global Change Biology REU Program	<i>May 2022 – August 2022</i>
Ecology, Evolution, and Marine Biology Department, UC Santa Barbara	

PRESENTATIONS

McDonald, A. M., Hofmann, G. E. (2023) Investigating Impacts Of Marine Heatwaves On A Primary Driver Of Kelp Forest Recovery And Decline: *Strongylocentrotus purpuratus*. Oral Presented at: Western Society of Naturalists Meeting; 2023 November 11; Monterey, California.

McDonald, A. M., (2020) What We Know: California Aquaculture and Environmental Stressors. Oral Presented at: Western Society of Naturalists Meeting; 2020 November 7; Virtual.

McDonald, A. M., Aluru, N., Krick, K. S., Karchner, S. I. (2020) Developmental Exposure to PCB153 (2,2',4,4',5,5'- Hexachlorobiphenyl) Alters Circadian Rhythms and the Expression of Clock and Metabolic Genes. Oral Presented at: Ocean Sciences Meeting; 2020 February 17; San Diego, California. Poster Presented at: WHOI Summer Undergraduate Research Forum; 2018 August 8; Woods Hole, Massachusetts.

McDonald A., Tekle Y. (2019) Vestigial genes? Examining the Presence of Genes Encoding for Radial Spoke and Central Pair in Non- flagellate Amoebas. Poster Presented at: Spelman College Research Day; 2019 April 25; Atlanta, Georgia. **Award: First Place**

McDonald A., Wang D. (2018). Analyzing Promoter 5' Deletions on InvINH1 during Seed Development. Poster Presented at: 1. Annual

Biomedical Research Conference for Minority Students; 2018 November 15; Indianapolis, Indiana.; 2. Spelman College Research Day; 2018 April 20; Atlanta, Georgia.

McDonald A., Bideau S., Gasca M. (2017) Urchins Are Important Herbivores and Mediate Positive Effects of Sediment on Algal Turfs. Oral Presented at: 1. University of California, Los Angeles Summer Research Symposium; 2017 August 18; Los Angeles, CA; 2. Spelman College Research Day; 2018 April 20; Atlanta, Georgia. Poster Presented at: Ocean Sciences Meeting; 2018 February 14; Portland, Oregon.

McDonald A., Lawrence A., Williams A. The Effect of Germline Mutations on the MSH2 Gene in Hereditary Nonpolyposis Colorectal Cancer. Poster session presented at: Spelman College Research Day; 2017 April 21; Atlanta, GA.

HONORS AND PROFESSIONAL AFFILIATIONS

Black Women in Ecology, Evolution, and Marine Science Member

Black in Marine Science Member

Association for the Sciences of Limnology and Oceanography

Phi Beta Kappa Honor Society

Beta Beta Beta National Biology Honor Society